Mechanical Timing Relays

1. Introduction to Mechanical Timing Relays

Mechanical timing relays are control devices used to delay the opening or closing of contacts in an electrical circuit. They allow automatic starting and stopping of operations at different time intervals.

Why Timing Relays Are Used

- To delay starting or stopping of machines
- To control sequential operations
- To ensure safe and orderly operation of industrial systems

2. Types of Contacts in Timing Relays

2.1 Instantaneous Contacts

- Change state immediately when the relay coil is energized
- No time delay involved
- Commonly used as holding or sealing contacts

2.2 Timed Contacts

- Change state only after a preset time delay
- Main functional part of timing relays

3. Methods of Time Delay

3.1 ON-Delay (Delay on Energize)

In an **ON-delay timer**, the time delay occurs when the relay coil is energized. The contacts change state only after the preset time has elapsed.

Example: Pressing a start button → motor starts after 10 seconds.

3.2 OFF-Delay (Delay on De-energize)

In an **OFF-delay timer**, the time delay occurs when the relay coil is de-energized. The contacts remain in their energized state for a certain time after power is removed.

Example: Switch OFF → fan continues running for 10 seconds → then stops.

4. ON-Delay Timer Circuits

4.1 NOTC: Normally Open, Timed Closed (ON-Delay)

Contact meaning: Normally open at rest, but **closes after a delay** when the timer coil is energized.

- **Coil OFF (initial):** Contact is OPEN ⇒ load is OFF.
- **Coil ON:** Timer starts counting immediately.
- **During timing:** Contact stays OPEN ⇒ load remains OFF.
- **After preset time:** Contact CLOSES ⇒ load turns ON.
- **Coil OFF again:** Contact OPENS instantly ⇒ load turns OFF immediately.

Step-by-step Example (Delayed motor start):
1) Press START pushbutton → timer coil energizes.
2) Motor still OFF for 10 s (waiting period).
3) After 10 s, NOTC contact closes → motor starts.
4) Press STOP → coil de-energizes → motor stops instantly.

4.2 NCTO: Normally Closed, Timed Open (ON-Delay)

Contact meaning: Normally closed at rest, but **opens after a delay** when the timer coil is energized.

- **Coil OFF (initial):** Contact is CLOSED ⇒ load is ON.
- **Coil ON:** Timer starts counting.
- **During timing:** Contact remains CLOSED ⇒ load stays ON.
- **After preset time:** Contact OPENS ⇒ load turns OFF.
- **Coil OFF again:** Contact CLOSES instantly ⇒ load turns ON immediately.

Step-by-step Example (Auto-stop alarm):
1) Alarm turns ON initially (contact closed).
2) When switch is activated, timer coil energizes and starts timing.
3) Alarm continues for 10 s.
4) After 10 s, NCTO contact opens → alarm turns OFF automatically.

5. OFF-Delay Timer Circuits

5.1 NOTO: Normally Open, Timed Open (OFF-Delay)

Contact meaning: Contact is normally open, **closes immediately when coil energizes**, and **opens after a delay when coil de-energizes**.

- **Coil OFF (initial):** Contact OPEN ⇒ load OFF.
- **Coil ON:** Contact CLOSES instantly ⇒ load ON immediately.
- **Coil OFF:** Timer starts counting (off-delay begins).
- **During off-delay:** Contact stays CLOSED ⇒ load remains ON.
- **After preset time:** Contact OPENS ⇒ load turns OFF.

Step-by-step Example (Fan run-on):
1) Turn ON switch → fan turns ON instantly.
2) Turn OFF switch → timer starts.
3) Fan keeps running for 10 s (cool-down).
4) After 10 s, NOTO contact opens → fan stops.

5.2 NCTC: Normally Closed, Timed Closed (OFF-Delay)

Contact meaning: Contact is normally closed, **opens immediately when coil energizes**, and **closes after a delay when coil de-energizes**.

- **Coil OFF (initial):** Contact CLOSED ⇒ load ON.
- **Coil ON:** Contact OPENS instantly ⇒ load OFF immediately.
- **Coil OFF:** Off-delay timing begins.
- **During off-delay:** Contact stays OPEN ⇒ load remains OFF.
- **After preset time:** Contact CLOSES ⇒ load turns ON again.

Step-by-step Example (Delayed restart):
1) System is ON initially (contact closed).
2) When control coil energizes, contact opens instantly → output turns OFF.
3) Coil de-energizes (power removed) → timer starts.
4) Output remains OFF for 10 s.
5) After 10 s, contact closes → output turns ON (delayed return).

6. Summary Comparison Table

Contact Type	Normal State	Timer Type	Delayed Action
NOTC	Open	ON-delay	Closes after energizing
NCTO	Closed	ON-delay	Opens after energizing
NOTO	Open	OFF-delay	Opens after de-energizing
NCTC	Closed	OFF-delay	Closes after de-energizing

Key Exam Points

- ON-delay $\rightarrow$ delay occurs when coil is energized
- OFF-delay $\rightarrow$ delay occurs when coil is de-energized
- Energizing action is instant in OFF-delay timers
- De-energizing action is instant in ON-delay timers